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/ **Practice Exam - 1**

Correct Answer

Practice Exam - 1

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A 26-year-old female patient is admitted with left upper quadrant pain and several weeks of black tarry stools. Four months prior, a Heartmate III LVAD was implanted. She has a past medical history significant for multiple gastrointestinal (GI) bleeds, a lower extremity deep vein thrombosis, left ventricular thrombosis, and hypertension. Upper endoscopy revealed a non-bleeding arteriovenous malformation in the upper body of the stomach as well as several areas of focal gastric erythema. Anticoagulation was reduced and an intravenous proton pump inhibitor started.

Which one of the following medications would be the best therapeutic option to reduce recurrent GI bleeding in this patient?



A. Octreotide



B. Ethinylestradiol

C. Thalidomide

D. Bevacizumab

Commentary

References

Testing Point: The Board-certified AHFTC specialist should know how to prevent common LVAD complications.

Correct Answer: A. Octreotide

Rationale: Gastrointestinal bleeding is a frequent complication of LVAD placement due to the continuous flow leading to dysregulation of angiogenesis-related mediators and shear stress. Arteriovenous malformations (AVM) are responsible for many cases (29-44%) of bleeding/rebleeding in LVAD patients, and are diffuse and multiple in up to 29%-44%. Endoscopy may be able to identify an acutely bleeding lesion, but one bleeding AVM lesion usually portends others along the GI tract that may bleed in the future, as well as the formation of new AVMs. Therefore, medical therapy targeted at the driving factors of these lesions has been used to reduce episodes of rebleeding and ultimately decrease the number of hospitalizations, endoscopic procedures and blood transfusions.

Octreotide (Answer A), a somatostatin analogue, is a common first agent used to reduce recurrent GI bleeding in LVAD patients following failure of conservative medical therapy. It is believed that octreotide reduces bleeding rates by decreasing portal venous pressures due to splenic vasodilation, enhancing platelet aggregation, and inhibition of angiogenesis in the GI tract through downregulation of VEGF and suppression of digestive enzymes. Octreotide is currently dosed 50 to 100 µg subcutaneous twice daily or octreotide long-acting release 20 to 30 mg intramuscularly monthly. Octreotide can be administered as an outpatient and can be found at most community pharmacies, making Answer A the best answer.

Hormone therapies, either estrogen-only or estrogen in combination with progesterone, have been studied in the prevention of AVM bleeds with conflicting results. Additionally, concerns for thromboembolism with estrogen-based therapies is a major concern in this already pro-thrombotic population. Based on these risks and the fact that our patient has a history of thromboembolic events, oral ethinylestradiol (Answer B) would not be an adequate option. Thalidomide (Answer C) in low doses can be antiangiogenic and could be a potential treatment option. Some studies describe an altered ratio of angiopoietin-1 to angiopoietin-2 as potentially contributing to the development of AVMs, and thalidomide may be helpful in restoring the ratio. However, due to its significant teratogenic side effects (even after one dose), thalidomide should be used with extreme caution in women of childbearing potential. Additionally, the drug carries a significant REMS program and must be ordered from a specialty pharmacy. Thus, in this patient, Answer C is not the most ideal choice. Answer D, bevacizumab, a humanized monoclonal antibody against VEGF, may lower the rate of GI bleeding in refractory cases. However, bevacizumab has been shown to cause ovarian failure in women of child bearing age and has been associated with venous thromboembolic events. Based these risks, Answer D would not be the best choice in this patient.

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